ABSTRACT

Convolutional neural networks, have become a powerful tool in medical image analysis, offering the opportunity to improve the accuracy of brain tumor detection by learning complex features from large and diverse datasets. This project focuses on developing an efficient brain tumor detection system using CNN architecture. The system takes MRI images as input and predicts the presence of brain tumors with high accuracy. This model is developed in three stages in the first stage, the dataset is pre-processed, enhancing its size and diversity. In the second stage pre-trained MobileNetV2 CNN architecture used for feature extraction, in the last stage neural network is trained for detection of the disease.

The detection of brain tumor model is implemented in python programming language by using the key libraries, such as TensorFlow and Keras, which are utilized for CNN model development. The frontend of the system is built accepting the Flask framework, enabling user interaction and facilitating the submission of MRI images for detection of tumor. The MobileNetV2 architecture is utilized for feature extraction, and the model is compiled with Adam optimizer and binary cross-entropy loss function.

To measure the performance of the system, extensive training and testing were conducted using a diverse dataset of MRI images, including both presence and absence of tumor's. The accuracy of the model was found to be approximately 94.95% after 15 epochs of training. The confusion matrix demonstrates the model's proficiency in distinguishing between positive and negative, with minimal false predictions.

Future enhancements could encompass incorporating more advanced architectures, exploring ensemble techniques for improved accuracy, or integrating real-time MRI image processing for swift predictions. Moreover, integrating a user-friendly interface through Flask could enhance accessibility and usability, promoting seamless interaction for medical practitioners and patients. The potential for real-time monitoring and collaboration with medical professionals for accurate diagnoses stands as a promising direction for further development.

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CHAPTER 1

1. INTRODUCTION

Deep neural network focuses on creating an advanced brain tumor detection model using TensorFlow and Keras. It begins with preprocessing the dataset to enhance diversity and size, crucial for robust training. Feature extraction utilizes MobileNetV2, a pre-trained architecture, followed by customization with a dense layer for binary detection of brain tumors presence or absence. The model is then trained over multiple epochs to optimize its predictive accuracy, evaluated using standard metrics. A practical test demonstrates the model's functionality, and its ability to save enhances future usability. Confusion matrix and visualizations further evaluate and represent the model's performance, resulting in a reliable brain tumor prediction model poised for real-world application. In proposed work showcases the detection for brain tumor model using cutting-edge machine learning techniques. Efficient feature extraction, thorough training, and comprehensive evaluation, ensuring the model's readiness for practical implementation in medical diagnostics, promising to enhance brain tumor diagnosis with accuracy and efficiency.

1.1 Problem Definition

Creating a robust algorithm to swiftly and accurately detect brain tumors from MRI scans is vital, given the potential life-threatening consequences of misinterpretation or delayed diagnosis. This imperative necessitates leveraging machine learning for automated detection. The process entails collecting diverse imaging datasets, preprocessing and extracting meaningful features, selecting appropriate machine learning models like convolutional neural networks (CNNs), training and validating the models, integrating them into clinical workflows, and continuously refining the algorithm for improved performance. The ultimate goal is to facilitate prompt and precise medical intervention through an efficient, automated brain tumor detection system

1.2 Purpose and Scope

The fundamental aim of this project is to engineer a highly reliable and efficient brain tumor detection system, utilizing advanced machine learning techniques. This initiative involves crafting a sophisticated model capable of analyzing diverse medical images, accurately identifying distinct brain tumor features, and offering valuable diagnostic insights. The overarching objective is to significantly assist medical professionals in their decision-making process, enabling them to make precise and prompt diagnoses. Ultimately, this technological advancement seeks to elevate the standard of patient care by enhancing the efficiency and accuracy of brain tumor detection in the realm of medical imaging.

1.3 Objectives

- To construct a machine learning model with a high degree of accuracy in detecting brain tumours from medical images.
- To enhance the model's accuracy and sensitivity to ensure that it can precisely identify subtle or complex tumors.
- To create a user-friendly tool that integrates seamlessly into the existing medical diagnostic workflow to expedite the decision-making process for healthcare professionals.

1.4 Target audience

The primary audience includes medical professionals, particularly radiologists and oncologists. They can use this tool as an assisting mechanism for their diagnoses. Additionally, the research community, specifically those involved in medical imaging and artificial intelligence, would benefit from this study.

1.5 Technologies Required

For the successful implementation of this project, the following technologies are essential:

- **Python:** Python is chosen as the primary programming language due to its vast array of libraries and tools conducive to machine learning.
- **TensorFlow and Keras:** TensorFlow and Keras, powerful machine learning frameworks, are vital for developing and training the neural network model.
- **Matplotlib:** Matplotlib, a popular data visualization library, is crucial for visually representing data and model performance.
- **TensorFlow Hub:** TensorFlow Hub, which provides a repository of pre-trained models, is utilized to leverage existing knowledge and optimize the model's performance.
- **Convolutional Neural Networks (CNNs):** Convolutional Neural Networks (CNNs), a class of deep learning models particularly effective for image analysis, constitute the core technique for accurate brain tumor predition. Integration of these technologies harmoniously forms the foundation for the successful realization of the project objectives.

CHAPTER 2

2. LITERATURE SURVEY

More number of paper available on brain tumor detection using neural networks such as ANN, BPNN. Few papers are reported on CNN model for prediction of medical anomalies. A model was developed by Shraddha S. More et al., [1] for brain tumor detection which have often relied on small datasets so they have augmented the dataset, highlighting the need for more extensive datasets and advanced algorithms such as Convolutional Neural Networks (CNNs), which have shown promising results with 87.42% training accuracy in this study.

Cinarer and Emoroglu in their paper [2], proposes a model for the classification of brain tumor by machine learning algorithms. This study assesses different machine learning algorithms for classifying MR brain image features, with SVM. demonstrating superior performance with a 90% accuracy rate compared to other methods. In the recent paper [3], researchers focused on the analysis of brain tumor MRI images, categorizing them into "normal" and "abnormal" states. They employed a Convolutional Neural Network (CNN) classifier, achieving a notable prediction accuracy of 93%. This approach highlighted the efficacy of CNNs in accurately distinguishing between normal and abnormal MRI images, showcasing their potential as a robust tool for precise brain tumor detection through medical imaging. The study's success in achieving a high prediction accuracy underscores the promising role of CNNs in enhancing diagnostic accuracy and expediting medical decision-making for patients with brain tumors.

A model was proposed by a group of researchers in [4], for brain tumor detection by using deep learning models, Convolutional Neural Networks (CNNs) and VGG-16 architecture, has emerged as a valuable tool for brain tumor diagnosis using MRI images. This study, based on a dataset of 253 patients, provides a comparative analysis of CNN and VGG-16. Brain tumor detection and segmentation image processing is challenging due to diversity and the potential for human-assisted errors. T. Hossain et al., in their paper [5] employs Fuzzy C-Means clustering and traditional classifiers like SVM, KNN, MLP, Logistic Regression, Naïve Bayes, and Random Forest, followed by a Convolutional Neural Network (CNN)

approach, this work achieving an impressive accuracy of 97.87% for distinguishing abnormal and normal pixels based on texture and statistical features. S. Kumar et al., in their work [6] developed a model for brain tumor detection using CNN, for both cancerous and non-cancerous. Their accurate detection via automated techniques on MRI images is vital and gives promising results with an accuracy of 90.2%. In [7] authors emphasize the pivotal role of Convolutional Neural Networks (CNNs) in brain tumor detection and segmentation. It shows improved performance and achieved good results through effective CNN utilization. The study delves into key parameters like optimization techniques, epoch count, batch size, and learning rate, strategically refining these aspects to enhance the overall efficiency of brain tumor detection and segmentation. This research underlines the significant potential ofCNNs in optimizing brain tumor diagnosis through meticulous parameter tuning.

A group of authors in their study [8] declare that traditional MRI brain tumor segmentation methods are more time-consuming limiting accessibility. This study presents the VGG-19, ResNet50, and DenseNet121 for brain tumor detection, reaching of 92% (ResNet50), 90% (DenseNet121), and 88% (VGG19). Spotting brain tumors in their initial stages is essential to avoid severe problems. In [9] Sai et al., employs Convolutional Neural Networks, which contains VGG16 and VGG-19 convolution layers. Both the architecture gives the results in the rate of 95% and 96%, respectively. A modified VGG-16 model with deconvolution and CRF-RNN also attains a commendable accuracy of 92.3%. In another paper [10],
S. B. Fayyad and A. A. Ibrahim use Convolutional Neural Network (CNN) for brain tumor detection and classification. Remarkably, they achieved exceptional precision rates of 98.29% during training and 98.029% during testing, highlighting the robustness of the CNN algorithm. Their approach involved a diverse dataset with images of various tumor types, showcasing the algorithm's proficiency in accurate classification across different tumor presentations. This underscores the potential of CNNs as a reliable tool for precise brain tumor diagnosis.

These research papers collectively underscore the crucial aspect of brain tumor detection and classification in medical research. Despite the frequent limitation of small datasets, researchers have cleverly utilized CNNs, often coupled with data augmentation techniques,

to overcome this constraint. This utilization has yielded highly promising outcomes, as evidenced by the attainment of high training accuracy, affirming the adaptability and effectiveness of CNNs in learning intricate features from medical imaging data. Moreover, extensive comparative analyses have been conducted, systematically evaluating various machine learning algorithms and consistently showcasing the superior performance ofCNN-based models in the precise classification of brain tumors. Recent studies have even demonstrated CNNs' prowess in accurately categorizing MRI brain images into normal or abnormal, a pivotal step in the diagnostic process. Furthermore, researchers are increasingly focusing on the fine-tuning of CNN parameters, optimizing architectures, and experimenting with diverse optimization techniques and learning rates to enhance their efficiency in detecting and segmenting brain tumors. These collective advancements firmly position CNNs as indispensable tools in the domain of brain tumor detection, holding immense potential to significantly enhance diagnostic accuracy and, consequently, contribute to timelyand precise medical interventions, ultimately leading to improved outcomes for patients.

CHAPTER 3

3. METHODOLOGY

The proposed project work consists of the following stages which are shown in the figure 3.1.

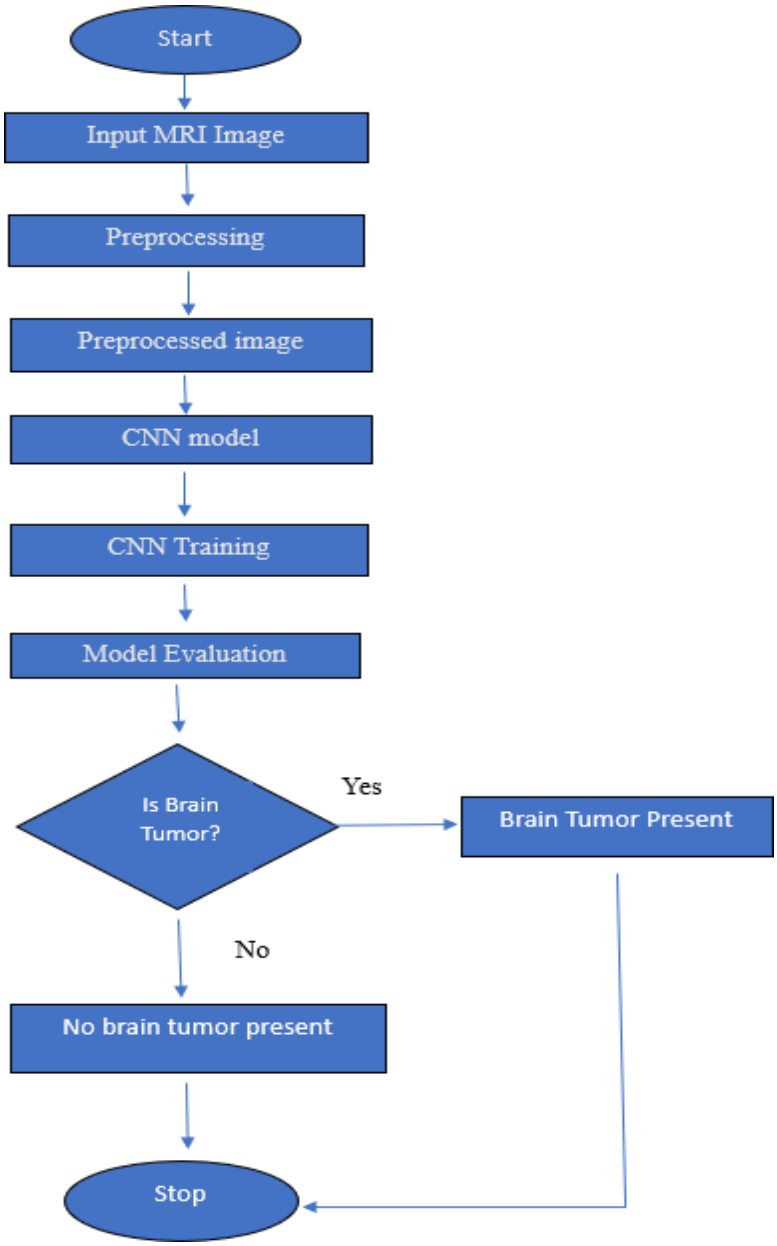

Figure 3.1: Work Flow of the proposed model.

The proposed model for detection of brain tumor approach using CNN (convolutional neural networks). In this approach, the system takes a JPG or PNG or JPEG-format MRI images as input and proceeds to identify any brain tumor within it or not. It involves several key steps. First, the dataset is preprocessed, enhancing its size and diversity. Next, the feature extraction is performed using MobileNetV2, a pre-trained architecture, which extracts meaningful features from the images. These features are then fed into a custom-built densely connected layer for binary detection, differentiating between the presence and absence of brain tumors. The model is compiled using the Adam optimizer and binary cross-entropy loss function, aiming to maximize predictive accuracy. The training is executed over 15 epochs, and validation data is used to assess the model's performance at each epoch. After training, accuracy metrics and a confusion matrix are computed to evaluate the model's effectiveness. Additionally, a practical test is conducted to demonstrate the model's ability to make predictions. Finally, the trained model is saved for future usage, consolidating the project's methodology into a comprehensive pipeline for accurate brain tumor detection.

3.1 Data Flow Diagrams

Level-0 DFD

In figure 3.2 shows that, the user interacts with the application interface, providing input in the form of MRI images. These images are then preprocessed to suit the Convolutional Neural Network (CNN) designed for brain tumor detection. The CNN processes the preprocessed images to produce output, indicating the presence or absence of a brain tumor, which is then presented to the user for interpretation and further action.

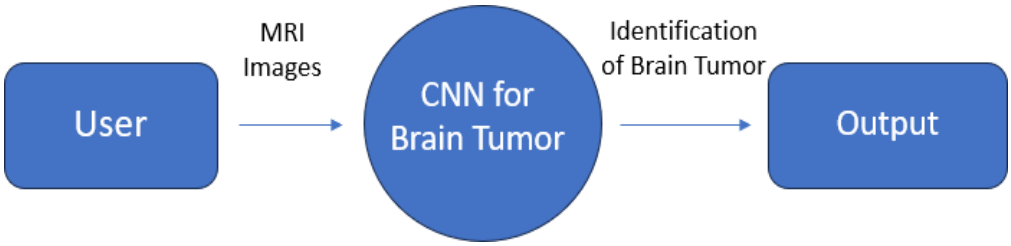

Figure 3.2: Context Data Flow for Brain Tumor Prediction using CNN.

Level-1 DFD

In figure 3.3 level 1 DFD for the brain tumor detection, external data sources such as MRI image datasets provide input for MRI image processing. The User Interface allows users to interact with the application. Input Image Storage handles the storage of input images. Image Preprocessing (block 1) involves preparing the input images for the CNN model. The CNN (block 2) processes the preprocessed images for brain tumor detection. Model Training and Evaluation (block 3) includes training the CNN model and evaluating its performance. Finally, Result Presentation involves displaying the brain tumor detection results to the users.

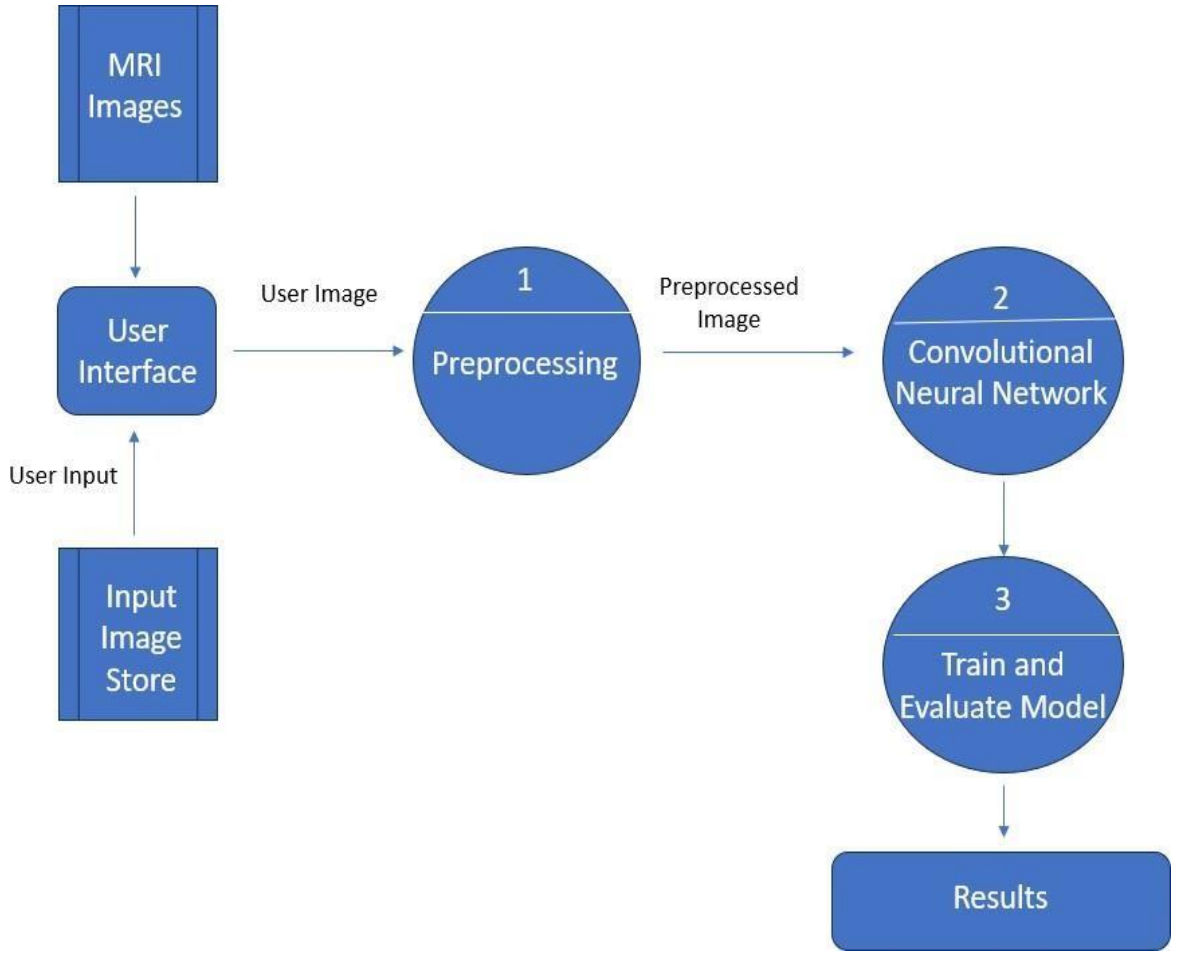

Figure 3.3: Process Flow for Brain Tumor Predication using CNN.

Level-2 DFD

The Convolutional Neural Network model designed for brain tumor detection follows a structured architecture involving convolutional layers (block 2.1) that extract intricate features from the input MRI images. These layers are complemented by pooling layers (block 2.2), strategically placed to reduce spatial dimensions, aiding in feature retention and computational efficiency. Fully connected layers (block 2.3) serve for detection, making decisions based on the learned features. Additionally, dropout layers (block 2.4) are incorporated to alleviate overfitting, enhancing the model's generalization capabilities. Ultimately, this CNN model produces an output, typically a probability or a binary detection, indicating the likelihood or presence of a brain tumor based on the learned features and the network's complex interconnectedness as showed in figure 3.4.

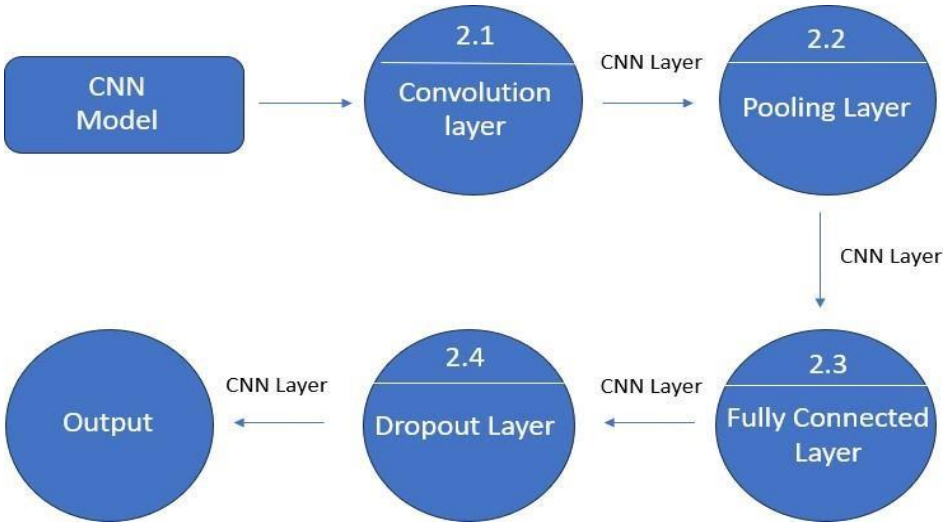

Figure 3.4: Exploration Data Flow Diagram for Brain Tumor Prediction using CNN.

3.2 Data collection and preprocessing

3.2.1 Data collection

About 253 brain tumor images were collected from publicly available data set and sorted into two distinct categories as images with tumor (155 images) and images without tumor (98 images) and finally developed image database as shown in the figure 3.5.

3.2.2 Data pre-processing

In this work data preprocessing is done in three stages. In the first stage, Resizing and Standardization: Resize the images to a consistent size such as 224x224 pixels to ensure uniformity. Standardize pixel values to be within the range 0 and 1 by dividing by 255. In the second stage, Label Encoding: Encode the categorical labels such as “Tumor and Non-Tumor” into numerical values (e.g., 0 and 1) for the model to understand. In the third stage, Train-Validation Split: Split the dataset into training and validation sets, ensuring a representative proportion of each class in both sets (e.g., using an 80-20 or 70-30 split).

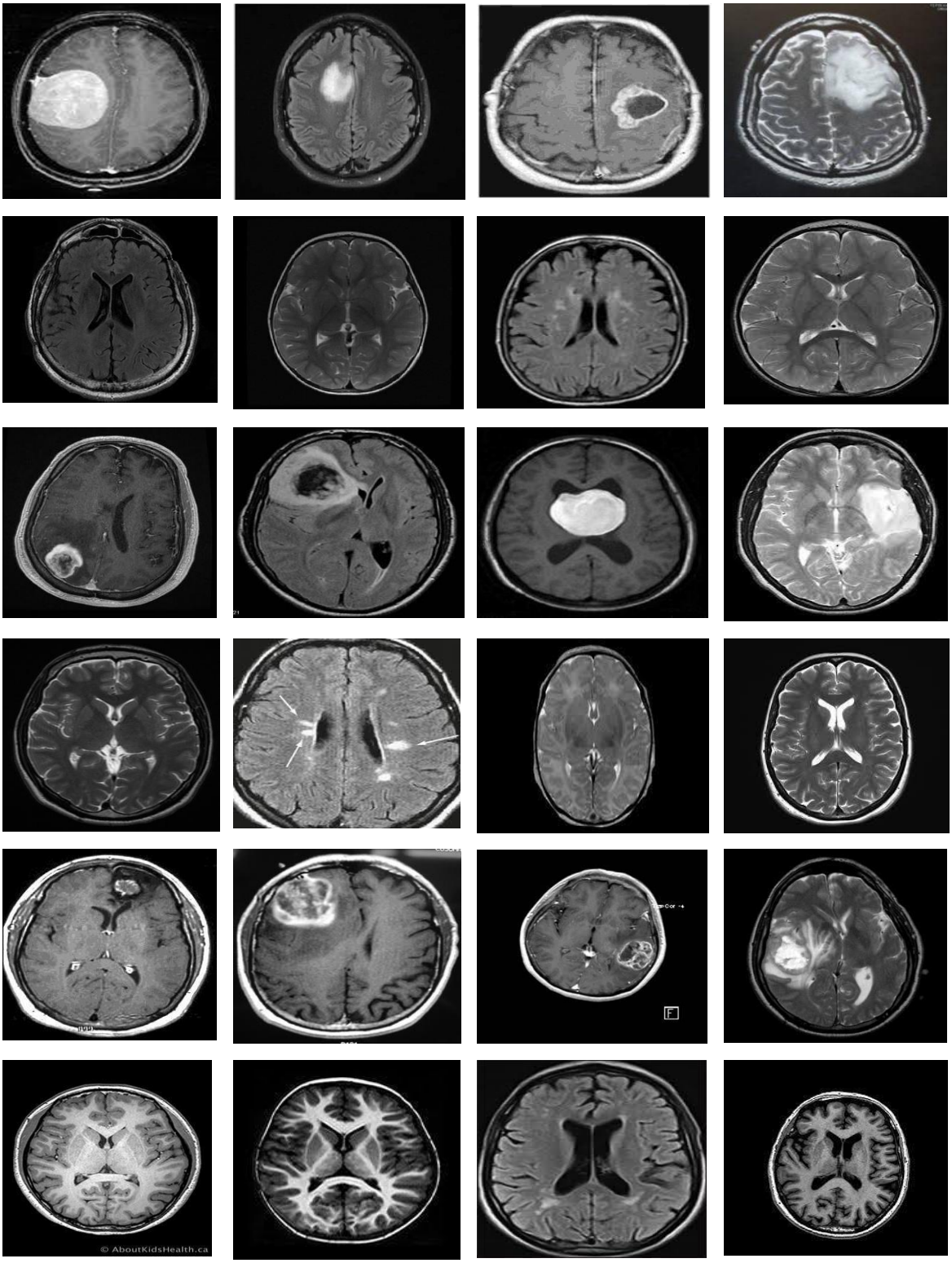

Figure 3.5: Sample of MRI brain images with and without tumor.

3.3 Convolutional Neural Network

Convolutional Neural Network (CNN) is a specialized type of computer program that's excellent at understanding and interpreting visual data like images and videos. CNN is a powerful tool but requires millions of labelled data points for training. It mimics how the human brain processes images, recognizing patterns and features to make sense of what it sees. CNNs are particularly good at tasks like identifying objects in pictures, recognizing handwritten letters, or even detecting emotions on faces in images. The process of a CNN involves passing the input image through multiple layers of convolutions and pooling, extracting increasingly complex features at each stage. CNNs use backpropagation to adjust the model's weights during training, minimizing the difference between predicted and actual outputs shown in figure 3.6. Overall, CNNs revolutionized image analysis tasks, enabling computers to interpret and understand visual data, with applications ranging from facial recognition to medical image diagnosis and autonomous driving.

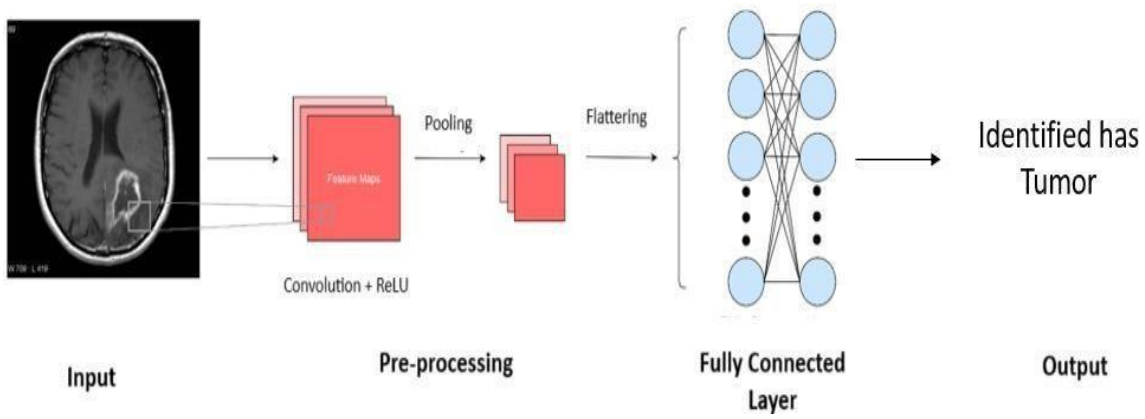

Figure 3.6: CNN architecture.

- **Input Layer:**
  The 244 x 244-pixel value of an MRI image represented as a 3D matrix is consider in this work as input.

- **Convolutional Layer**

  Convolutional layers apply filters (kernels) to input images to extract features. These filters help identify patterns like edges, shapes, textures in the images. In this project, the MobileNetV2 architecture is used, which inherently has convolutional layers to extract features from images.

- **Activation Function**

  The ReLU (Rectified Linear Unit) activation function is a popular choice for prediction. It squashes the network's output between 0 and 1, making it suitable for predictions, where the goal is to predict the given input into either “no brain tumor" (0) and "brain tumor" (1).

- **Pooling Layer**

  The Pooling Layer is a fundamental component in Convolutional Neural Networks (CNNs) used for analysing and processing image data. Its primary purpose is to reduce the spatial dimensions of each feature map while retaining important information. This reduction in spatial dimensions helps in decreasing the computational load and mitigating overfitting, making the model more efficient and easier to train.

  Types of Pooling Layer

  - **Max Pooling:** Max Pooling is the most common type of pooling used in CNNs. It operates on each window (often 2x2) traversing the input feature map and outputs the maximum value from that window.

  - **Average Pooling:** In Average Pooling, instead of selecting the maximum value from the window, it calculates the average value of the window and uses that for down sampling the feature map.

- **Fully Connected Layer**

  Fully connected layers connect every neuron in one layer to every neuron in the next layer. These layers are often used after the convolutional and pooling layers to learn high-level features. In this project, there is a single dense (fully connected) layer with 1 neuron and a ReLU activation function.

- **Output Layer**

  The Output Layer generates predictions based on the features learned by the previous layers. For binary detection, a single neuron is used with a ReLU activation function, providing a probability of the image belonging to a particular class (0 or 1).

3.4 Model Training

- **Compiling the Model:** The model is compiled using the adam optimizer, which is effective for gradient-based optimization. Binary cross-entropy loss function is chosen since this is a binary detection problem. Accuracy is chosen as the evaluation metric.
- **Training:** The model is trained using the training set and validated using a validation set. The training is performed for 15 epochs, with the model adjusting its weights to minimize the loss and improve accuracy.

3.5 Model Evaluation

- **Validation Evaluation:** Evaluate the model using the validation set to assess its performance in terms of Training and validation accuracy and loss are monitored and plotted to visualize the model's learning progress.
- **Fine-Tuning:** If needed, fine-tune the model by adjusting hyperparameters, layers, or even using additional techniques like learning rate scheduling.

3.6 Single Prediction and Model Saving

- **Prediction:** A single test image is loaded and pre-processed for prediction using the trained model. The model predicts whether the test image contains a brain tumor or not based on the computed probability.
- **Model Saving:** Save the trained model architecture, weights, and optimizer configuration to file for future use without retraining.

3.7 Confusion Matrix and Visualization

- **Compute Confusion matrix:** The model's performance is evaluated using a confusion matrix based on a separate test set or the validation set. The confusion matrix provides insights into the true positive, true negative, false positive, and false negative predictions.
- **Plot Performance Metrics:** The project concludes by visualizing the training and validation accuracy and loss using plots, providing insights into the model's performance during training.

CHAPTER 4

4. RESULTS AND DISCUSSION

The proposed model involves building a CNN (convolutional neural network) using transfer learning with MobileNetV2 to detect brain tumor images into 'no tumor' and 'tumor' categories. The model achieves an impressive 94.95% validation accuracy after 15 epochs of training on a dataset of 253 images. A sample test image is correctly predicted, showcasing the model's efficacy. The generated confusion matrix demonstrates the model's overall strong performance in distinguishing between tumor and non-tumor. This project underscores the potential of deep learning in medical image detection, specifically in aiding brain tumor identification.

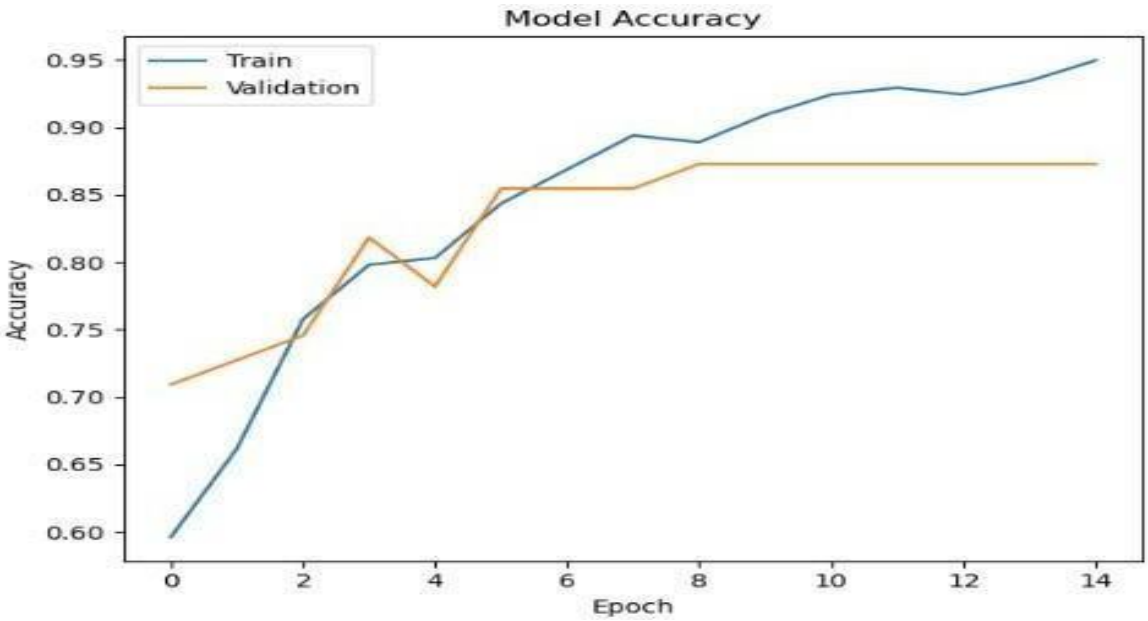

Figure 4.1: Epoch model accuracy.

The model achieves an impressive 94.95% validation accuracy after 15 epochs of training on a dataset of 253 images. In figure 4.1 the image shows a line graph of model accuracy over the course of training. The x-axis represents the epoch, which is the number of times the model has seen the entire training dataset. The y-axis represents the accuracy, which is the percentage of training examples that the model correctly predicted.

In the line graph the blue line representing the training accuracy and the yellow line representing the validation accuracy. Training accuracy is the model's accuracy on the dataset it was trained on, while validation accuracy is the model's accuracy on a separate validation dataset that it hasn't seen during training.

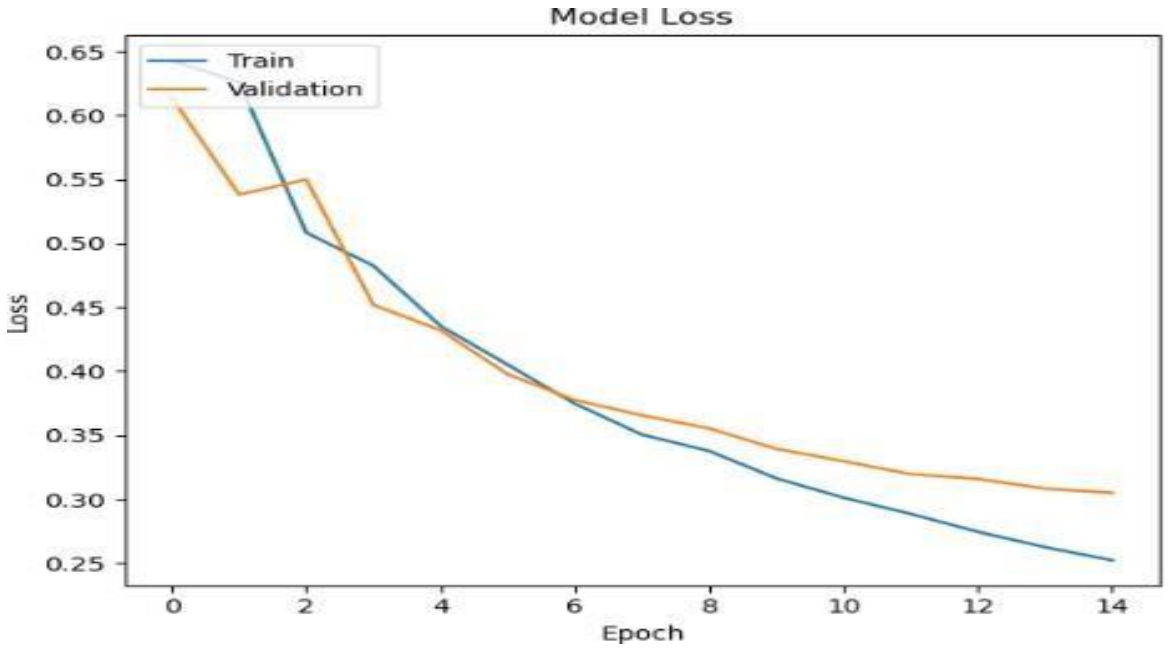

Figure 4.2: Epoch model loss.

The loss of the model, as measured by the binary cross-entropy loss function, decreases over each epoch during training showed in figure 4.2. The x-axis represents the epoch, which is the number of times the model has seen the entire training dataset. The y-axis represents the loss, which is a measure of how well the model's predictions match the actual outputs.

In the line graph one is blue line for the training loss and one is yellow line for the validation loss. The training loss is the loss of the model on the training dataset, while the validation loss is the loss of the model on a held-out validation dataset. The training loss typically starts out high and decreases as the model is trained. The validation loss usually follows the same trend, but it may lag behind the training loss. This is because the validation dataset is not used to train the model, so it is a more challenging test of the model's performance.

```
Found 253 images belonging to 2 classes.
8/8 [==============================] - 13s 2s/step
Confusion Matrix:
[[ 85  13]
 [  5 150]]
```

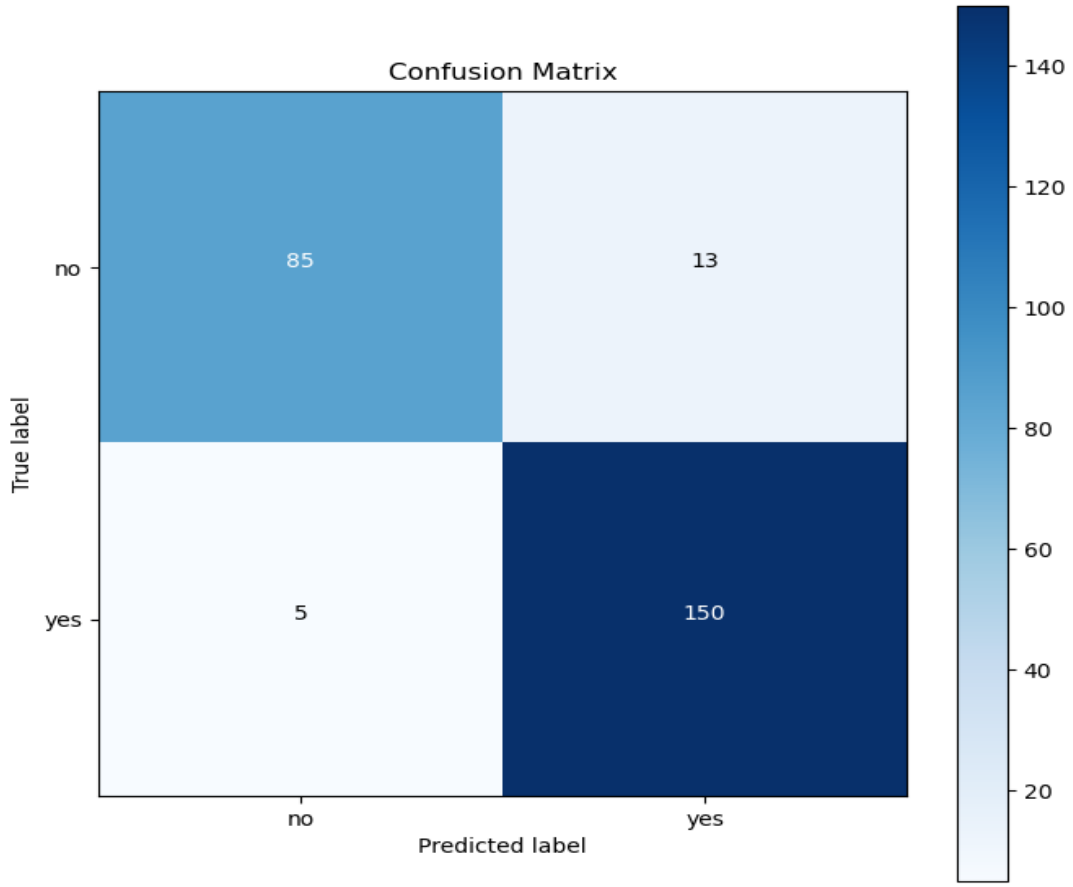

Figure 4.3: Confusion matrix.

Figure 4.3 presents a confusion matrix, a vital tool for assessing a detection model's performance by comparing predicted and actual outcomes. The matrix has two rows and two columns. The rows represent the actual class labels, and the columns represent the predicted class labels. In the context of tumor detection, this matrix aids in evaluating the model's ability to distinguish between tumor-positive and tumor-negative cases. The matrix comprises four key components: True Positives, True Negative, False Positives and False Negatives.

- A session with the system.

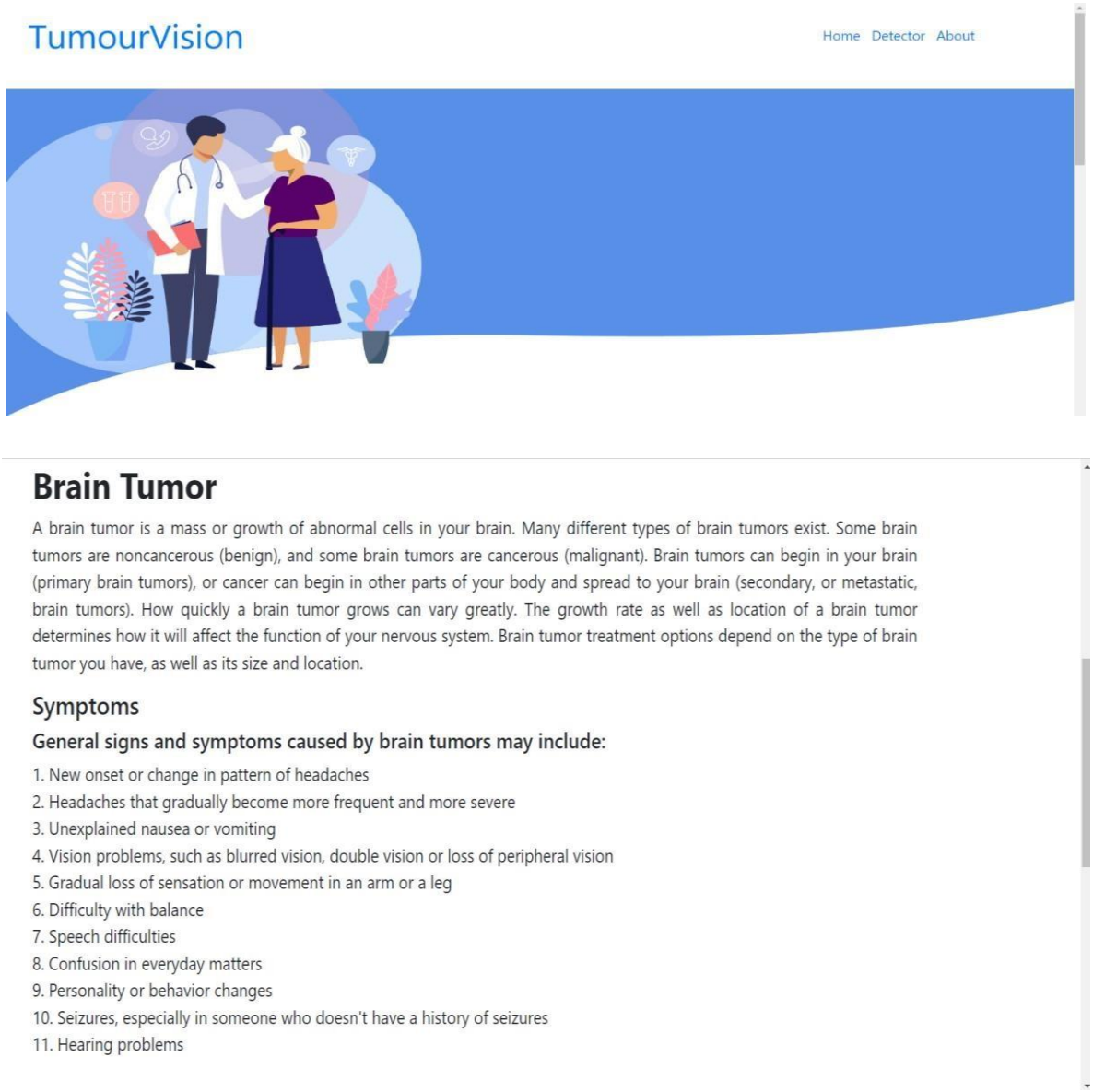

Figure 4.4: Home page of the project.

Homepage of the "Brain Tumor Prediction" is shown in the figure 4.4 which serves as the initial landing point, providing essential information about brain tumors. The homepage interface provides comprehensive information on brain tumors, encompassing details about their characteristics, and associated diagnostics.

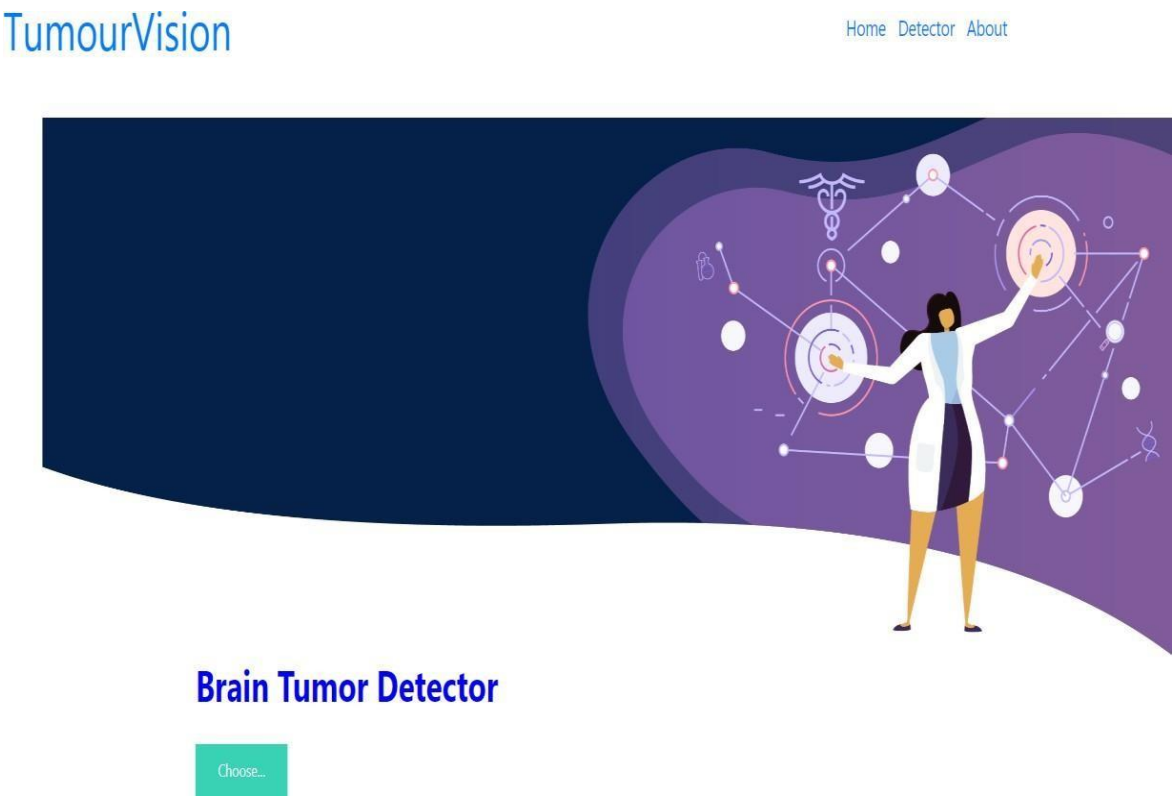

Figure 4.5: Brain tumor detector page.

In figure 4.5 the brain tumor detector page, allows users to upload their MRI images, enabling a quick and automated analysis to detect the presence of brain tumors. This feature aims to provide a user-friendly interface for early identification of potential brain abnormalities. By clicking the "Choose" button, users can select an MRI image from their device, initiating the upload process, and triggering subsequent analysis to identify possible brain tumors.

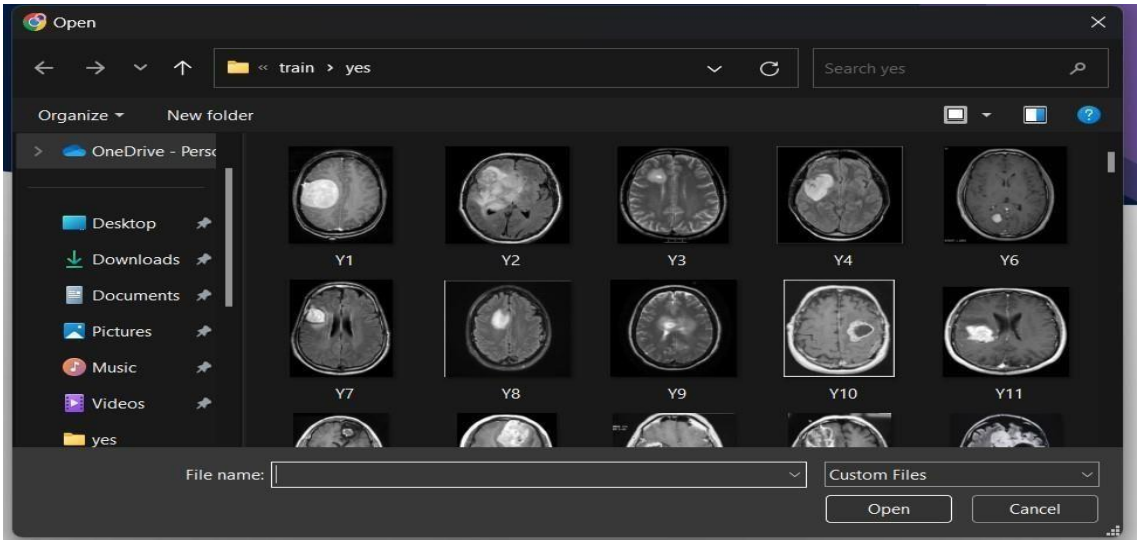

Figure 4.6: Image Database for selecting input image.

In figure 4.6 the image database section allows users to browse and select an input MRI image from a MRI image dataset. Users can explore various MRI images to choose the most suitable input for the brain tumor detection process, enhancing the flexibility of this project.

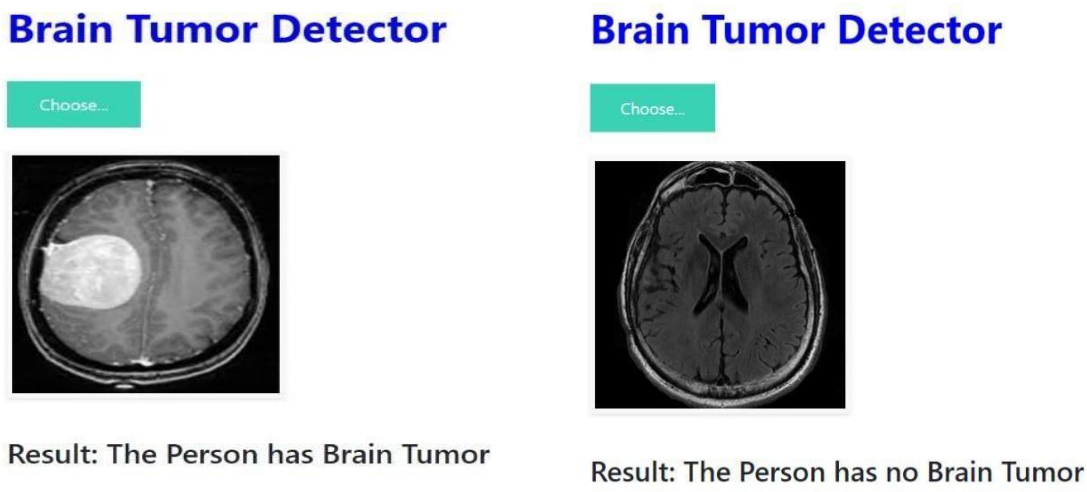

Figure 4.7: Brain tumor detector.

Figure 4.7 displays the results of the brain tumor prediction. It showcases a visual representation of the analysis, indicating the presence or absence of a brain tumor based on the uploaded MRI image.

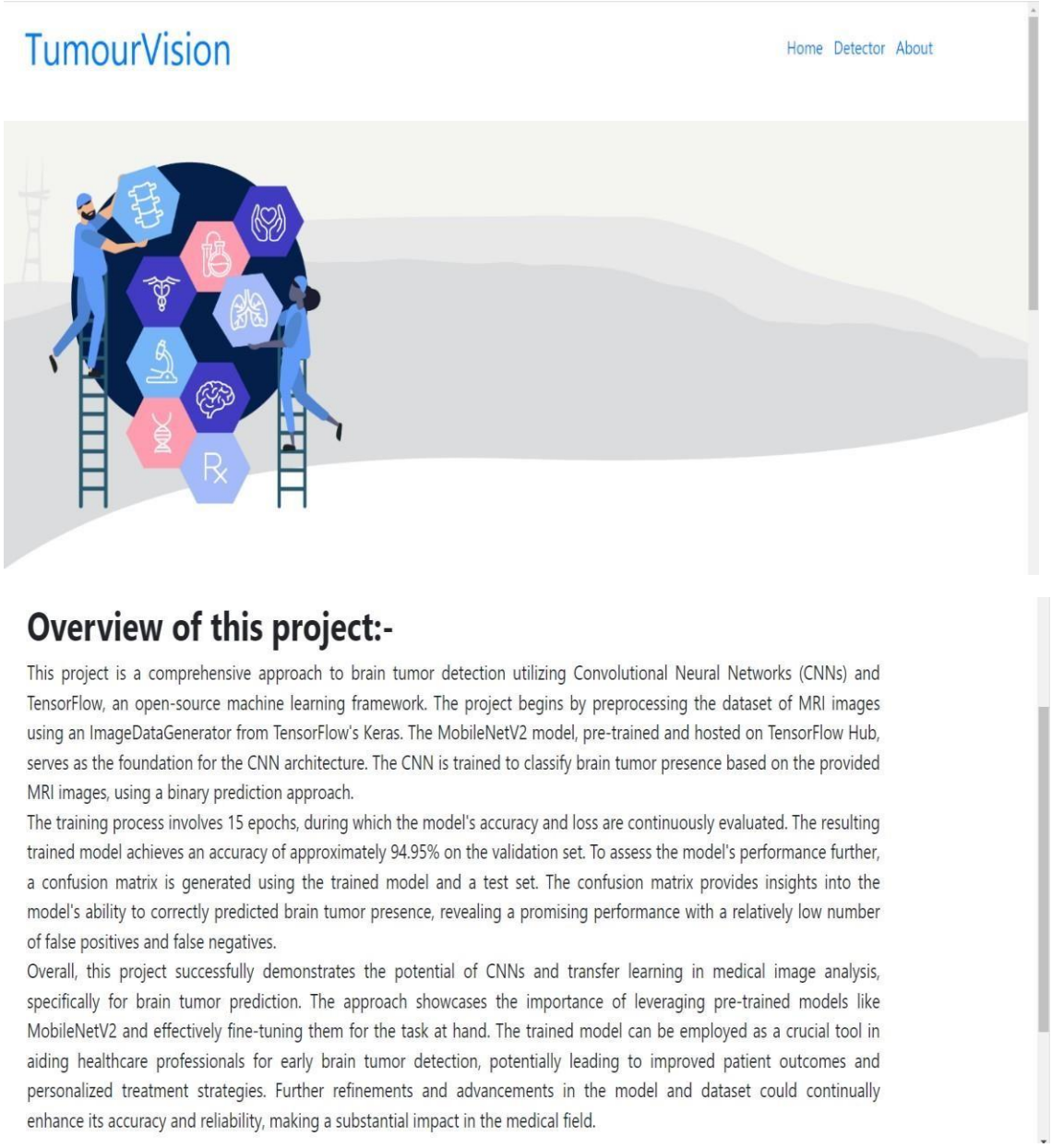

Figure 4.8: Overview of this project.

In figure 4.8 talks about the "About" page of the project which offers a comprehensive overview, detailing the project's objectives and scope. It provides insights into the motivation behind the project, its goals, and the intended impact in the field of brain tumor detection.

CHAPTER 5

5. CONCLUSION AND FUTURE ENHANCEMENT

5.1 Conclusion

A robust brain tumor detection system by utilizing a convolutional neural network (CNN) architecture, specifically MobileNetV2, to predict the presence of brain tumors in MRI images with a high degree of accuracy. Through a three-stage approach involving data preprocessing, feature extraction using a pre-trained CNN, and training a neural network for binary detection, the model achieves an accuracy of approximately 94.95% after 15 epochs of training. While the system demonstrates promising results, there are limitations to be acknowledged, including the potential for false positives/negatives due to image variations and the necessity for a diverse and extensive dataset for more robust model training. Addressing these limitations requires a continual focus on dataset improvement and exploring advanced architectures to enhance the model's accuracy and reliability.

5.2 Future Enhancement

This project aims to advance brain tumor detection through several key strategies. First, the integration of more advanced architectures, such as ensembles of CNNs, is being explored to significantly enhance accuracy. Leveraging the collective intelligence of multiple CNN models can lead to more robust predictions and a better understanding of complex patterns in MRI images. Secondly, a focus on real-time MRI image processing is critical to enable swift and immediate predictions. This enhancement would greatly improve the clinical utility of the tool, allowing for rapid decision-making and timely interventions.

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